

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2020

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2020 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $3 + 4 \times 6 - 4$. [2]

2. Expand $(x + 1)(x + 1)$. [3]

3. Overnight in Gweru the temperature was -3°C . By 14:00 it was 32°C . Find the rise. [2]

4. In a class of Gifford High, $n(A) = 13$ (play soccer), $n(A \cap B) = 4$ (play both). How many play soccer only? [2]

5. Simplify $3^3 \times 3^2$. Leave as a power of 3. [2]

6. A map of Gweru uses scale 3 : 800000. A road is 12 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm $\leftrightarrow$ 1 km if 3:8 is first simplified — instead: scale 3 cm to 8 km.) [3]

7. A tank in Mkoba holds 42 litres. It is $\frac{2}{6}$ full. How many litres are in the tank? [3]

8. Find 15% of 40. [3]

- 9.** Find the HCF and LCM of 48 and 20. **[4]**
- 10.** Write 90000 in standard form. **[3]**
- 11.** Factorise $x^2 - 9$. **[4]**
- 12.** Area of a circle radius 7 cm. Take $\pi = 22/7$. **[4]**
- 13.** A trader at Mkoba buys a bag of potatoes for \$80 and sells for \$88. Percentage profit? **[4]**
- 14.** A cylindrical drum has radius 7 cm and height 4 cm. Find its volume. $\pi = 22/7$. **[4]**
- 15.** Find the gradient of the line joining (2, 3) and (4, 11). **[3]**
- 16.** Factorise $x^2 + 5x + 6$. **[4]**
- 17.** Find the midpoint of (2, 6) and (2, 2). **[3]**
- 18.** A regular polygon has 9 sides. Find each interior angle. **[4]**
- 19.** A sequence is 7, 11, 15, 19, ... Find an expression for the n th term. **[4]**
- 20.** Find the mean of 12, 8, 7, 6, 9. **[3]**

- 21.** In right-angled triangle PQR, $\hat{P} = 30^\circ$ and hypotenuse PR = 10 cm. Find PQ, opposite 30° . **[4]**
- 22.** Make r the subject of $C = 2\pi r$. **[4]**
- 23.** A bag contains 7 red, 6 blue and 5 green beads. $P(\text{not green})$? **[3]**
- 24.** Work out $(4 \frac{1}{2} ; 4 \frac{2}{3})(5 ; 2)$. **[4]**
- 25.** A right-angled triangle has shorter sides 10 cm and 24 cm. Find the hypotenuse. **[4]**
- 26.** Solve $4x + 2 < 18$. **[3]**
- 27.** Solve $3(x + 2) = 18$. **[4]**
- 28.** $AB \parallel CD$. A transversal makes an acute angle of 66° with AB. Find the co-interior angle on CD. **[3]**
- 29.** Solve $\begin{cases} x + y = 7 \\ x - y = 1 \end{cases}$. **[4]**
- 30.** A triangular plot has base 6 m and perpendicular height 11 m. Find its area. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).